HEPATIC ADENOMA — CLINICAL PROTOCOL BOUNDARY ADDENDUM V7

Companion to the existing standard/guidance-aligned protocol, not a relabeling as high risk. The original clinical protocol is unchanged. These OPEN boundary recommendations require ratification before being incorporated into clinical use.

Version 7 | Revised 2026-10-24 06:55 CDT | Institutional approval pending

POLICY AND ALLOCATION STATUS

Analyst A structured critical review through 24 October 2026. All 27 B decisions remain OPEN. Primary evidence, selected-recipient outcomes, pathway-level outcomes, guideline wording and local safeguards are distinguished. This is not a PRISMA-complete systematic review, meta-analysis, fresh external-analyst consensus or institutional approval.

Policy posting checked 24 October 2026: OPTN effective 1 October 2026; oncology guidance December 2025. Clinical suitability, allocation access, donor protection and research authorization remain separate. Recheck before any application. [1] [2]

Use the adjacent study tables before interpreting a proposed cutoff. Abstract-only extraction and missing full PDFs are labeled; missing outcome estimates are not inferred from graphs or from one minus disease-free survival. Linked full text is in this document's own PDF depot when available.

HCA-B1 — WHICH OF THE FOUR GUIDANCE ROUTES IS ACTUALLY BEING INVOKED?

HCA-B1 — WHICH OF THE FOUR GUIDANCE ROUTES IS ACTUALLY BEING INVOKED?

QUESTION AND COMPARISON

Which exception-review route applies to this unresectable adenoma phenotype, and does that route correspond to a defensible clinical benefit from LT?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
NLRB oncology guidance [2] Current NLRB hepatic-adenoma guidance. Primary source | Full-text PDF Locator: December 2025 oncology guidance, hepatic adenoma p11. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** GSD/Abernethy; OR unresectable β-catenin-mutated HCA; OR unresectable adenomatosis >10 lesions; OR unresectable, nonoperative-unresponsive and progressive/complicated HCA. **Do not infer:** These are review routes, not proof of benefit or a mandatory transplant recommendation. Current text does not require all four conditions simultaneously or cirrhosis for all routes.

Study • population • source	Results • interpretation boundary
ACG 2024 [3] ACG 2024, HCA recommendations/key concepts 11–18. Primary source | Full-text PDF Locator: Table 3 key concepts 11–18 and HCA management text. Review depth: Complete local PDF reviewed; HCA/EHE key concepts and supporting text; key concepts are not GRADE-rated recommendations	**Reported result:** Stable women with lesions ≥5 cm: risk-factor change and 6–12-month observation, resection if not <5 cm. Men/β-catenin lesions warrant earlier definitive review; LT may be considered in selected unresectable/complicated disease. **Do not infer:** Six versus 6–12 months is a real timing nuance. ACG adenomatosis definition is ≥10, whereas current NLRB route says >10; definitions cannot silently replace each other.
European Association for the Study of the Liver 2016 [4] EASL 2016 benign liver tumor guideline. Primary source | Full-text PDF Locator: HCA recommendations and multiple-HCA discussion, PDF pp9–10. Review depth: Primary full text reviewed; local complete PDF available	**Reported result:** Six-month observation in suitable women; ≥5 cm or growth prompts resection. LT is not routine for multiple adenomas; underlying liver disease can alter consideration. **Do not infer:** Older clinical guidance is narrower than current NLRB alternative exception routes. A resection threshold is not an LT threshold.
Nault 2017 [5] Molecular study: 533 adenomas from 411 patients across 28 centers. Primary source | Full-text PDF Locator: Abstract; molecular classification and clinicopathological Results; pp880–894. Review depth: Complete local PDF reviewed: molecular cohort, subtype definitions and outcomes	**Reported result:** Symptomatic bleeding in 14% of patients; 3% developed HCC. Strong exon-3 CTNNB1 activation tracks malignant risk; exon-7/8 activation is biologically different; sonic-hedgehog subtype associates with bleeding. **Do not infer:** Observed prevalence is not individual lifetime risk. Molecular risk supports definitive-treatment assessment, not automatic LT or equivalence of all β-catenin variants.
Ziogas 2022 [6] UNOS 1987–2020: 199 listed; 142 transplanted, including 118 listed for HCA and 24 incidental diagnoses. Primary source | Full-text PDF Locator: Methods; UNOS Results pp3–5 and Figure 2; not the review-only survival curve. Review depth: Complete local PDF reviewed: Methods, UNOS transplant/listing cohorts and systematic-review limitations	**Reported result:** Crude waitlist mortality 9.0%. Post-LT 1/3/5-year OS 94.2%/89.7%/86.3%; 5-year CI 78.5–91.5%; median follow-up 62.9 months. **Do not infer:** No contemporary non-LT comparator; registry indication misclassification. Separate systematic-review patients must not be pooled with UNOS or Chiche as independent evidence.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Allocation-route eligibility and clinical necessity must be recorded separately. Lesion number, size and molecular subtype identify different questions. The NLRB four-route OR structure is more permissive than older clinical LT guidance; clinical risk still needs confirmation and alternatives review. [2] [3] [4] [5] [6]

- Adenomatosis is ≥10 in ACG terminology but >10 in the cited NLRB route: exactly ten lesions cannot be silently rounded into >10. Molecular subtypes are not biologically interchangeable. Selected-registry survival does not quantify the benefit of replacing an otherwise functioning liver.

Certainty / applicability: High confidence in the meaning of the current guidance text; low certainty for phenotype-specific LT benefit; moderate biological evidence for subtype heterogeneity, not a validated LT threshold.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- >10: literal allocation-guidance wording (at least 11); different from ACG's diagnostic definition.
- β-catenin: guidance category; genotype/phenotype should inform clinical risk, without silently rewriting the allocation category.
- Need for LT: multidisciplinary clinical decision beyond exception eligibility.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
NLRB oncology guidance [2] "Candidates may qualify for an exception, if they meet one of the following criteria:" December 2025 oncology guidance p11, hepatic adenoma Source | Full-text PDF	The routes are alternatives; route four has three conjunctive elements.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use the four NLRB routes as OR alternatives, then separately prove a clinical indication for LT. [2] [3] [4] [5] [6]

- Document at least one actual guidance route: (1) adenoma with GSD or Abernethy; OR (2) unresectable β-catenin-mutated adenoma; OR (3) unresectable adenoma with adenomatosis >10 lesions (at least 11); OR (4) unresectable AND unresponsive to nonoperative therapy AND progressive or complicated disease.
- Do not require all four routes, and do not impose cirrhosis as a national prerequisite.
  - Conversely, meeting one guidance route does not itself justify surgery.
- Require specialist pathology/radiology and an alternatives review before requesting an exception; proven malignant transformation must be evaluated under the appropriate cancer pathway.

Why retain or change it: Retain the four alternative NLRB routes and add explicit molecular subtype/lesion-count reconciliation. Do not invent a requirement for all four routes or convert the β-catenin label alone into an indication for LT.

INFORMATION REQUIRED BEFORE THE DECISION

Subtype/β-catenin result where needed; lesion count and distribution; GSD/shunt status; resectability; prior nonoperative treatment and complication record. Record the exact lesion count and, when available, the CTNNB1 exon/activation phenotype; exactly ten is not >10.

Clinical versus allocation: NLRB wording concerns exception review and remains nonbinding. Submission, eligibility and approval are distinct.

Roadmap / protocol reconciliation: Keep the benign-liver-tumor roadmap's conservative, resection and embolization options. This record does not change the standard HCA protocol.

Meeting record: HCA-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

HCA-B2 — HAS THE BENEFIT OF LT BEEN DISTINGUISHED FROM CONCERN ABOUT MALIGNANCY?

HCA-B2 — HAS THE BENEFIT OF LT BEEN DISTINGUISHED FROM CONCERN ABOUT MALIGNANCY?

QUESTION AND COMPARISON

When does LT offer enough incremental benefit over observation, resection or embolization to justify recipient and possible donor risk?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Chiche 2016 [7] European retrospective adenomatosis LT series, n=49, 1986–2013. Primary source | Full-text PDF Locator: Indications, explant pathology and patient-outcome Results; published pp516–526. Review depth: Primary full text reviewed; local complete PDF available	**Reported result:** Only 1/15 suspected-HCC cases had HCC confirmed. Three deaths within 90 days; five retransplants. HCC recurrence in 3/17 with malignant explants, all fatal; 41/49 alive and well at final report. **Do not infer:** Selected surgical series, not untreated transformation risk; overlaps European cases in later reviews. Low diagnostic yield warns against using suspicion alone.
Ziogas 2022 [6] UNOS 1987–2020: 199 listed; 142 transplanted, including 118 listed for HCA and 24 incidental diagnoses. Primary source | Full-text PDF Locator: Methods; UNOS Results pp3–5 and Figure 2; not the review-only survival curve. Review depth: Complete local PDF reviewed: Methods, UNOS transplant/listing cohorts and systematic-review limitations	**Reported result:** Crude waitlist mortality 9.0%. Post-LT 1/3/5-year OS 94.2%/89.7%/86.3%; 5-year CI 78.5–91.5%; median follow-up 62.9 months. **Do not infer:** No contemporary non-LT comparator; registry indication misclassification. Separate systematic-review patients must not be pooled with UNOS or Chiche as independent evidence.

Study • population • source	Results • interpretation boundary
Haring 2019 [8] Retrospective hormone-withdrawal cohort, n=78. Primary source | TMC: retrieve PDF Locator: Published/institutional abstract Results; DOI 10.1111/liv.14074. Review depth: Primary abstract and institutional repository abstract verified; PDF pending; do not claim complete-text review	**Reported result:** At median 1.3 years after withdrawal, regression >30% 37.2%, complete regression 5.1%, stable 56.4%, progression 1.3%; no HCA-related complications reported. **Do not infer:** Primary abstract/repository extraction; not a transplant comparison. Follow-up exceeds six months, so it does not prove that six months is a biological deadline.
van Rosmalen 2019 [9] Six-center retrospective embolization cohort, n=59; 23 bleeding and 36 nonbleeding. Primary source | Full-text PDF Locator: Methods; Safety section pp1364–1365; size-response Results/Figure 1. Review depth: Complete local PDF reviewed: Methods and safety/size-response Results	**Reported result:** Six major complications (10%); median diameter 48 mm at median 5.5-month imaging. One death at three months after emergency embolization with hemorrhagic shock; no clinically overt rebleeding of treated lesions. **Do not infer:** No randomized comparator; embolization is neither harmless nor proven to eliminate malignant potential. Requires lesion-specific technical and pathology assessment.
European Association for the Study of the Liver 2016 [4] EASL 2016 benign liver tumor guideline. Primary source | Full-text PDF Locator: HCA recommendations and multiple-HCA discussion, PDF pp9–10. Review depth: Primary full text reviewed; local complete PDF available	**Reported result:** Six-month observation in suitable women; ≥5 cm or growth prompts resection. LT is not routine for multiple adenomas; underlying liver disease can alter consideration. **Do not infer:** Older clinical guidance is narrower than current NLRB alternative exception routes. A resection threshold is not an LT threshold.
ACG 2024 [3] ACG 2024, HCA recommendations/key concepts 11–18. Primary source | Full-text PDF Locator: Table 3 key concepts 11–18 and HCA management text. Review depth: Complete local PDF reviewed; HCA/EHE key concepts and supporting text; key concepts are not GRADE-rated recommendations	**Reported result:** Stable women with lesions ≥5 cm: risk-factor change and 6–12-month observation, resection if not <5 cm. Men/β-catenin lesions warrant earlier definitive review; LT may be considered in selected unresectable/complicated disease. **Do not infer:** Six versus 6–12 months is a real timing nuance. ACG adenomatosis definition is ≥10, whereas current NLRB route says >10; definitions cannot silently replace each other.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

LT is feasible in carefully selected adenoma patients, but the strongest comparative gap is benefit versus less invasive care. Overdiagnosed malignant transformation, real perioperative/retransplant burden and frequent regression after hormone withdrawal argue against transplantation for concern alone. [7] [6] [8] [9] [4] [3]

- Some patients have unresectable β-catenin lesions, underlying metabolic/vascular disease, recurrent hemorrhage or uncontrolled progression. For them, waiting or repeated local therapy may be harmful. A six-month reassessment is not a reason to postpone urgent definitive treatment or automatically transplant a stable lesion.

Certainty / applicability: Low certainty for selected LT prognosis; very low for comparative LT benefit. Nonrandomized alternative-treatment data support individualized sequencing, not a universal delay.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- Six months: EASL reassessment point; ACG allows 6–12 months in selected women. Not a mandatory waiting period for men, hemorrhage or malignant transformation.
- ≥5 cm/growth: trigger for definitive-treatment/resection assessment, not automatic LT.
- Two independent reviews: local anti-overtreatment safeguard, not a trial-validated number.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
European Association for the Study of the Liver 2016 [4] "6 months observation after lifestyle change" EASL 2016 guideline PDF p9, HCA recommendations Source | Full-text PDF	Applies to women in the relevant clinical setting, not a mandatory delay for every adenoma.
European Association for the Study of the Liver 2016 [4] "more than 10 lesions and underlying liver disease" EASL 2016 guideline PDF p10, multiple HCA discussion Source | Full-text PDF	A narrower clinical LT framing than the NLRB alternative exception routes.
Chiche 2016 [7] "LT should remain an extremely rare treatment for adenomatosis." Published PDF p2, abstract conclusion Source | Full-text PDF	The 49-patient series also recorded three operative deaths and five retransplantations; transplantation is not risk-free prophylaxis.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Keep LT exceptional; size or concern about malignancy alone is not a transplant indication. [4] [7] [6] [8] [9] [3]

- For suitable stable women, reassess after six months of hormone/lifestyle optimization; consider continued observation toward twelve months if the lesion is regressing and delay is clinically safe.
  - A lesion ≥5 cm or continued growth triggers resection/other definitive-treatment review, not automatic LT.
- Do not impose that waiting period on hemorrhage, high-risk male/β-catenin disease or confirmed malignant transformation when earlier treatment is appropriate.
- Proposed LT gate: two independent specialist reviews confirm that resection is not feasible and nonoperative options are unsuitable or have failed, plus a documented clinical-benefit basis: progressive/complicated disease, underlying liver-disease risk, or a high-risk molecular/malignant-transformation phenotype.
  - Stable unresectable high-risk exon-3 disease is not automatically excluded.
  - Pure diagnostic anxiety or an allocation route without a defined benefit is insufficient.

Why retain or change it: Refine the OPEN rule: reassess suitable stable women at six months, with continued observation toward twelve months when regressing and clinically safe (ACG). Preserve prompt review for high-risk/complicated disease and two independent alternatives reviews before proposed LT.

INFORMATION REQUIRED BEFORE THE DECISION

Subtype and malignancy review; sex/hormonal history; serial size; complications; embolization/resection feasibility; hepatic reserve; documented expected benefit and alternatives.

Clinical versus allocation: An exception is an allocation mechanism, not proof that expected benefit exceeds donor/graft risk.

Roadmap / protocol reconciliation: Retain the HCA roadmap's sex/subtype-sensitive treatment sequence. Confirmed cancer leaves the benign pathway.

Meeting record: HCA-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

Evidence review record: Five analyst comments and reconciliation | Prospective outcomes dataset

REFERENCES

[1] Organ Procurement and Transplantation Network. OPTN Policies: liver allocation and living donation. HRSA/OPTN. 2026. Official source | Full text PDF

Policy or professional guidance, not a primary treatment study. Official full PDF checked.

[2] Organ Procurement and Transplantation Network. Adult MELD Exceptions for Transplant Oncology. HRSA/OPTN. 2025. Official source | Full text PDF

Allocation review guidance. Complete local primary PDF available; relevant methods, selection, results and limitations reviewed.

[3] Frenette, Catherine; Mendiratta-Lala, Mishal; Salgia, Reena; et al. ACG Clinical Guideline: Focal Liver Lesions. American Journal of Gastroenterology. 2024;119(7):1235-1271. DOI | Full text PDF

Clinical research / guidance. Complete local PDF reviewed; HCA/EHE key concepts and supporting text; key concepts are not GRADE-rated recommendations.

[4] European Association for the Study of the Liver. EASL Clinical Practice Guidelines on the management of benign liver tumours. Journal of Hepatology. 2016;65(2):386-398. DOI | Full text PDF

Clinical practice guideline. Primary full text reviewed; local complete PDF available.

[5] Nault, Jean-Charles; Couchy, Gabrielle; Balabaud, Charles; et al. Molecular Classification of Hepatocellular Adenoma Associates With Risk Factors, Bleeding, and Malignant Transformation. Gastroenterology. 2017;152(4):880-894.e6. DOI | Full text PDF

Clinical research / guidance. Complete local PDF reviewed: molecular cohort, subtype definitions and outcomes.

[6] Ziogas, Ioannis A.; Tasoudis, Panagiotis T.; Serifis, Nikolaos; et al. Liver Transplantation for Hepatic Adenoma: A UNOS Database Analysis and Systematic Review of the Literature. Transplantation Direct. 2022;8(2):e1264. DOI | Full text PDF

Clinical research / guidance. Complete local PDF reviewed: Methods, UNOS transplant/listing cohorts and systematic-review limitations.

[7] Chiche, Laurence; David, Anaelle; Adam, René; et al. Liver transplantation for adenomatosis: European experience. Liver Transplantation. 2016;22(4):516-526. DOI | Full text PDF

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[8] Haring, Martijn P. D.; Gouw, Annette S. H.; de Haas, Robbert J.; et al. The effect of oral contraceptive pill cessation on hepatocellular adenoma diameter: A retrospective cohort study. Liver International. 2019;39(5):905-913. DOI | Local PDF pending

Clinical research / guidance. Primary abstract and institutional repository abstract verified; PDF pending; do not claim complete-text review.

[9] van Rosmalen, B V; Klompenhouwer, A J; de Graeff, J Jaap; et al. Safety and efficacy of transarterial embolization of hepatocellular adenomas. British Journal of Surgery. 2019;106(10):1362-1371. DOI | Full text PDF

Clinical research / guidance. Complete local PDF reviewed: Methods and safety/size-response Results.

REVISION NOTE

Revised 2026-10-24 06:55 CDT. Evidence-before-recommendation review of all 27 B questions. Earlier protocol versions and external analyst records are preserved; this is an Analyst A synthesis, not a new external-analyst vote.

Problem: HEHE/HCA B questions required the same evidence depth as the six high-risk protocols.
Correction and reason: Added a blue-format clinical boundary addendum with the full decision-specific evidence, uncertainty and OPEN proposals, without overwriting or relabeling the standard pathway.

Approval record Clinical lead _____ Allocation review _____ Research authorization if required _____ Donor review if applicable _____ Date _____